Slick Trace investigation report — case_0006

Generated 2026-09-26T12:21:52.565055+00:00 · reproducible investigation report

What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

Finding

No vessel in this scene can account for the observed slick. No candidate release reproduced the observation above the absolute fit gate. The system returns no name rather than the least-bad one; a source vessel may have left the scene before acquisition.

Scene

Field	Value
Incident	case_0006
Acquired (UTC)	2026-09-10T04:30:00+00:00
Sensor	synthetic SAR-like (VV proxy, single channel)
Data status	synthetic
Pixel spacing	433.59 m
Label source	generator ground truth (synthetic) — not a human or independently confirmed label

Evidence layers

Layer	Provenance
AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.

Traffic filtering

State	Vessels
excluded	1
candidate	4

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

Rank fragility

Leader retained in 1.0 of 80 tested perturbations. No tested perturbation changed the leader.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

File integrity

File	SHA-256 (first 16)
attribution_result.json	5db2e4757712bf85
coverage_result.json	2a86fa8542fd2367
forcing.npz	07595f69999ce4c5
forecast_result.json	92efc88a87e99f34
fragility_result.json	4811a6875cef91c3
historical_mapping.json	f600ca374180a6c5
manifest.json	2297dfa436dd8ea5
mask_groundtruth.png	57f9574e078cb2c6
naive_baseline_result.json	ea0683fbf3354ad0
next_observation_result.json	d5654d2e3535eb49
scene.png	e81930a0301fbc5e

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

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